Econ 101A
Section 8 (Solutions)

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Everything so far has been abstract: two goods x and y. This section puts the same machinery to work on two real decisions. How many hours to work, and how much to consume today versus tomorrow. Nothing new in the math. Each is a constrained utility maximization with a cleverly written budget. The only trick is reading the budget correctly and knowing what plays the role of a price.

1 Labor Supply: Consumption versus Leisure

You have a fixed time endowment T (hours in a day, say). Split it between leisure ℓ and work $h = T - \ell$. Consumption c has price 1, the wage is w, and you have non-labor income $M \geq 0$.

The budget. You spend on consumption what you earn plus your non-labor income:

$$c = M + wh = M + w(T - \ell) \quad \Longleftrightarrow \quad \underbrace{c + w\ell}_{\text{spending}} = \underbrace{M + wT}_{\text{full income}}.$$

Read the right-hand form as an ordinary budget. **The wage w is the price of leisure**: every hour you do not work costs you w in forgone earnings. The right side, $M + wT$, is "full income," the value of your time if you sold all of it, plus non-labor income.

Interior optimum. Maximize $u(c, \ell)$ subject to $c + w\ell = M + wT$. Same tangency as always:

$$\frac{u_\ell}{u_c} = w.$$

The rate at which you trade leisure for consumption equals the wage.

The no-work corner. If the interior solution wants $\ell > T$, that is infeasible: you cannot enjoy more leisure than you have. Then $\ell^* = T$, $h^* = 0$, $c^* = M$. There is a **reservation wage**: below it, work is not worth it and hours are zero.

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Comparative statics on hours. Two clean signs and one ambiguous one.

- $\partial h^*/\partial M < 0$: more non-labor income, if leisure is normal, means you buy more leisure and work less.
- $\partial h^*/\partial w$ is **ambiguous**. A higher wage raises the price of leisure (substitution effect: work more) but also makes you richer (income effect: buy more leisure, work less). When the income effect wins at high wages you get a **backward-bending** labor supply curve.

2 Intertemporal Choice: Today versus Tomorrow

Two periods, $t = 0, 1$. Incomes M_0, M_1. Consumption c_0, c_1, each priced at 1 in its own period. You can save or borrow at rate r.

The budget. Push everything to present value. A dollar tomorrow is worth $1/(1+r)$ today, so

$$c_0 + \frac{c_1}{1+r} = M_0 + \frac{M_1}{1+r} \equiv W.$$

The price of future consumption is $1/(1+r)$: to consume one more unit tomorrow you must set aside $1/(1+r)$ today. W is lifetime wealth in present value.

Preferences and the Euler equation. Utility is $U(c_0) + \dfrac{1}{1+\delta}U(c_1)$, where $\delta \geq 0$ is the rate of time preference (impatience) and $U' > 0$, $U'' < 0$. The Lagrangian gives $U'(c_0) = \lambda$ and $\dfrac{U'(c_1)}{1+\delta} = \dfrac{\lambda}{1+r}$. Divide to eliminate λ:

$$\boxed{U'(c_0) = \frac{1+r}{1+\delta}\,U'(c_1)} \qquad \text{(the Euler equation).}$$

It balances the marginal utility of a dollar spent today against a dollar saved and spent tomorrow.

The benchmark $r = \delta$. Then $U'(c_0) = U'(c_1)$, and since U' is strictly decreasing, $c_0 = c_1$: **perfect consumption smoothing**. Saving is $s = M_0 - c_0$. You save when income is front-loaded and borrow when it is back-loaded, to flatten the path.

3 Exercises

Exercise 1 (labor supply)

Let $u(c, \ell) = \ln c + \ln \ell$, with time endowment T, wage $w > 0$, and non-labor income $M \geq 0$.

(a) Write the budget in full-income form and set up the tangency condition.

(b) Solve for c^*, ℓ^*, and hours h^*.

(c) For which wages does the worker supply positive hours? Interpret the threshold.

(d) Sign $\partial h^*/\partial M$ and $\partial h^*/\partial w$. Is labor supply backward-bending here?

Solution. (a) Budget and tangency.
Start from consumption equals earnings plus non-labor income:

$$c = M + w(T - \ell).$$

Expand the right side: $c = M + wT - w\ell$. Move the leisure term to the left:

$$c + w\ell = M + wT \equiv F.$$

So leisure has price w and full income is $F = M + wT$.
Now the tangency condition. Compute the marginal utilities of $u = \ln c + \ln \ell$:

$$u_c = \frac{\partial}{\partial c}\big(\ln c + \ln \ell\big) = \frac{1}{c}, \qquad u_\ell = \frac{\partial}{\partial \ell}\big(\ln c + \ln \ell\big) = \frac{1}{\ell}.$$

The tangency condition $u_\ell/u_c =$ (price of leisure)/(price of c) $= w/1$ becomes

$$\frac{1/\ell}{1/c} = w \implies \frac{c}{\ell} = w \implies c = w\ell.$$

(b) Solve.
Substitute $c = w\ell$ into the budget $c + w\ell = F$:

$$w\ell + w\ell = F \implies 2w\ell = F \implies \ell^* = \frac{F}{2w}.$$

Then $c^* = w\ell^* = w \cdot \dfrac{F}{2w} = \dfrac{F}{2}$. Writing $F = M + wT$:

$$c^* = \frac{M + wT}{2}, \qquad \ell^* = \frac{M + wT}{2w}.$$

Hours are $h^* = T - \ell^*$. Put over a common denominator $2w$:

$$h^* = T - \frac{M + wT}{2w} = \frac{2wT - (M + wT)}{2w} = \frac{2wT - M - wT}{2w} = \frac{wT - M}{2w} = \frac{T}{2} - \frac{M}{2w}.$$

(c) Positive hours.
We need $h^* > 0$. Since the denominator $2w > 0$, the sign is the sign of the numerator:

$$h^* > 0 \iff wT - M > 0 \iff wT > M \iff w > \frac{M}{T}.$$

The reservation wage is M/T. If the wage is below the value your own time already has through non-labor income, you do not work: you sit at the corner $\ell^* = T$, $h^* = 0$, $c^* = M$.

(d) Comparative statics.
Use the tidy form $h^* = \dfrac{T}{2} - \dfrac{M}{2w}$.
Vary M (treat w as constant): $\dfrac{\partial h^*}{\partial M} = -\dfrac{1}{2w} < 0$. More unearned income, less work.

Vary w (treat M as constant): write the second term as $-\frac{M}{2}\,w^{-1}$. Its derivative is $-\frac{M}{2}\cdot(-1)w^{-2} = \frac{M}{2w^2}$. So

$$\frac{\partial h^*}{\partial w} = \frac{M}{2w^2} > 0.$$

Hours rise with the wage. With Cobb–Douglas the substitution effect always dominates, so labor supply slopes up and is *not* backward-bending. Backward-bending needs preferences where the income effect can outweigh substitution at high wages.

Exercise 2 (intertemporal choice, log utility)

Let $U(c) = \ln c$, so lifetime utility is $\ln c_0 + \frac{1}{1+\delta}\ln c_1$, with budget $c_0 + \frac{c_1}{1+r} = W$ where $W = M_0 + \frac{M_1}{1+r}$.

(a) Write the Euler equation and solve for c_0^* and c_1^*.

(b) Evaluate at $M_0 = 100$, $M_1 = 50$, $r = \delta = 0$. Find consumption and saving. Is she a saver or borrower, and why?

(c) Holding r fixed, sign $\partial c_0^*/\partial\delta$ and interpret.

Solution. (a) Euler equation.
Set up the Lagrangian with the present-value budget:

$$\mathcal{L} = \ln c_0 + \frac{1}{1+\delta}\ln c_1 + \lambda\Big(W - c_0 - \frac{c_1}{1+r}\Big).$$

First-order conditions:

$$\frac{\partial\mathcal{L}}{\partial c_0} = \frac{1}{c_0} - \lambda = 0 \;\Rightarrow\; \lambda = \frac{1}{c_0}, \qquad \frac{\partial\mathcal{L}}{\partial c_1} = \frac{1}{1+\delta}\frac{1}{c_1} - \frac{\lambda}{1+r} = 0.$$

Put $\lambda = 1/c_0$ into the second condition:

$$\frac{1}{1+\delta}\frac{1}{c_1} = \frac{1}{(1+r)c_0}.$$

Cross-multiply (multiply both sides by $(1+r)c_0c_1(1+\delta)$, or just rearrange): $(1+r)c_0 = (1+\delta)c_1$, hence the Euler equation

$$c_1 = \frac{1+r}{1+\delta}\,c_0.$$

Now use the budget. Substitute this c_1 into $c_0 + \frac{c_1}{1+r} = W$:

$$c_0 + \frac{1}{1+r}\cdot\frac{1+r}{1+\delta}c_0 = W \;\Longrightarrow\; c_0 + \frac{c_0}{1+\delta} = W.$$

Factor c_0 and combine the bracket over a common denominator:

$$c_0\Big(1 + \frac{1}{1+\delta}\Big) = W \;\Longrightarrow\; c_0\cdot\frac{(1+\delta)+1}{1+\delta} = W \;\Longrightarrow\; c_0\cdot\frac{2+\delta}{1+\delta} = W.$$

Solve:

$$c_0^* = W\,\frac{1+\delta}{2+\delta}.$$

Then $c_1^* = \dfrac{1+r}{1+\delta}c_0^* = \dfrac{1+r}{1+\delta}\cdot W\dfrac{1+\delta}{2+\delta} = \dfrac{(1+r)W}{2+\delta}$.

(b) Numbers.

With $r=\delta=0$: first the wealth, $W = M_0 + \dfrac{M_1}{1+r} = 100 + \dfrac{50}{1} = 150$. Then

$$c_0^* = 150\cdot\frac{1+0}{2+0} = \frac{150}{2} = 75, \qquad c_1^* = \frac{(1+0)\cdot 150}{2+0} = 75.$$

Saving is $s = M_0 - c_0^* = 100 - 75 = 25 > 0$. She **saves** 25. Her income is front-loaded ($M_0 = 100 > M_1 = 50$), and with $r=\delta$ she wants a flat path $c_0 = c_1$, so she moves 25 from period 0 into period 1.

(c) Effect of impatience.

We have $c_0^* = W\dfrac{1+\delta}{2+\delta}$, and $W = M_0 + M_1/(1+r)$ does not contain δ, so treat W as a constant. Differentiate $\dfrac{1+\delta}{2+\delta}$ by the quotient rule, $(f/g)' = (f'g - fg')/g^2$ with $f = 1+\delta$, $g = 2+\delta$, $f' = g' = 1$:

$$\frac{d}{d\delta}\frac{1+\delta}{2+\delta} = \frac{(1)(2+\delta) - (1+\delta)(1)}{(2+\delta)^2} = \frac{2+\delta-1-\delta}{(2+\delta)^2} = \frac{1}{(2+\delta)^2}.$$

Therefore

$$\frac{\partial c_0^*}{\partial \delta} = \frac{W}{(2+\delta)^2} > 0.$$

More impatience raises consumption today and so lowers saving. A higher δ discounts the future more, so she pulls resources toward the present.

Exercise 3 (intertemporal comparative statics with the IFT)

Keep general U with $U' > 0$, $U'' < 0$, and utility $U(c_0) + \dfrac{1}{1+\delta}U(c_1)$. Substitute the budget as $c_1 = M_1 + (1+r)(M_0 - c_0)$, so saving is $s = M_0 - c_0$.

(a) Write the optimality condition as $F(c_0; r) = 0$ and confirm $F_{c_0} \neq 0$.

(b) Use the implicit function theorem to sign $\partial c_0^*/\partial r$ for a *borrower* ($s < 0$).

(c) Explain why the sign is ambiguous for a *saver* ($s > 0$).

Solution. (a) Set up F and compute F_{c_0}.

Start from the Euler equation $U'(c_0) = \dfrac{1+r}{1+\delta}U'(c_1)$. Substitute $c_1 = M_1 + (1+r)(M_0 - c_0)$ and move everything to one side:

$$F(c_0; r) = U'(c_0) - \frac{1+r}{1+\delta}\,U'\big(M_1 + (1+r)(M_0 - c_0)\big) = 0.$$

Differentiate with respect to c_0, term by term. First term: $\dfrac{d}{dc_0}U'(c_0) = U''(c_0)$. Second term: it is a constant $\dfrac{1+r}{1+\delta}$ times $U'(c_1)$, and c_1 depends on c_0, so use the chain rule. First find how c_1 moves with c_0:

$$\frac{\partial c_1}{\partial c_0} = \frac{\partial}{\partial c_0}\big[M_1 + (1+r)(M_0 - c_0)\big] = (1+r)\cdot(-1) = -(1+r).$$

So the derivative of the second term is $-\dfrac{1+r}{1+\delta}U''(c_1)\cdot\big(-(1+r)\big) = +\dfrac{(1+r)^2}{1+\delta}U''(c_1)$. Adding,

$$F_{c_0} = U''(c_0) + \frac{(1+r)^2}{1+\delta}U''(c_1).$$

Both $U''(c_0) < 0$ and $U''(c_1) < 0$, so $F_{c_0} < 0$. In particular $F_{c_0} \neq 0$, so the IFT applies.

(b) Sign for a borrower.

The IFT gives $\dfrac{\partial c_0^*}{\partial r} = -\dfrac{F_r}{F_{c_0}}$. Compute F_r by differentiating F with respect to r, holding c_0 fixed. The first term $U'(c_0)$ has no r, so it drops. The second term is a product of $\dfrac{1+r}{1+\delta}$ and $U'(c_1)$, both depending on r, so use the product rule.

Piece one: $\dfrac{d}{dr}\dfrac{1+r}{1+\delta} = \dfrac{1}{1+\delta}$, times $U'(c_1)$, gives $\dfrac{U'(c_1)}{1+\delta}$.

Piece two: $\dfrac{1+r}{1+\delta}$ times $\dfrac{d}{dr}U'(c_1) = U''(c_1)\cdot\dfrac{\partial c_1}{\partial r}$. Here $\dfrac{\partial c_1}{\partial r} = \dfrac{\partial}{\partial r}\big[M_1 + (1+r)(M_0 - c_0)\big] = (M_0 - c_0) = s$. So piece two is $\dfrac{1+r}{1+\delta}U''(c_1)\,s$.

Because the second term of F carries a minus sign,

$$F_r = -\Big[\frac{U'(c_1)}{1+\delta} + \frac{1+r}{1+\delta}U''(c_1)\,s\Big] = -\frac{1}{1+\delta}\Big[U'(c_1) + (1+r)U''(c_1)\,s\Big].$$

Now the sign. Since $F_{c_0} < 0$, dividing gives $\dfrac{\partial c_0^*}{\partial r} = -\dfrac{F_r}{F_{c_0}} = \dfrac{F_r}{|F_{c_0}|}$, so it has the *same sign as* F_r.

For a borrower, $s < 0$. Look at the bracket $U'(c_1) + (1+r)U''(c_1)s$: the second piece is $(1+r) > 0$ times $U''(c_1) < 0$ times $s < 0$, which is positive. So the bracket is (positive $U'(c_1)$) + (positive) = positive, and $F_r = -\frac{1}{1+\delta}$(positive) < 0. Hence $\dfrac{\partial c_0^*}{\partial r} < 0$: a higher rate makes borrowing costlier and makes the borrower poorer, and both push today's consumption down.

(c) Why the saver is ambiguous.

For a saver, $s > 0$. Now the second piece $(1+r)U''(c_1)s$ is $(+)(-)(+)$ = negative. The bracket becomes (positive $U'(c_1)$) + (negative), whose sign depends on the sizes, i.e. on the curvature of U. Economically: substitution pushes c_0 down (saving now pays more), but the saver is richer (income effect pushes c_0 up), and either can win.